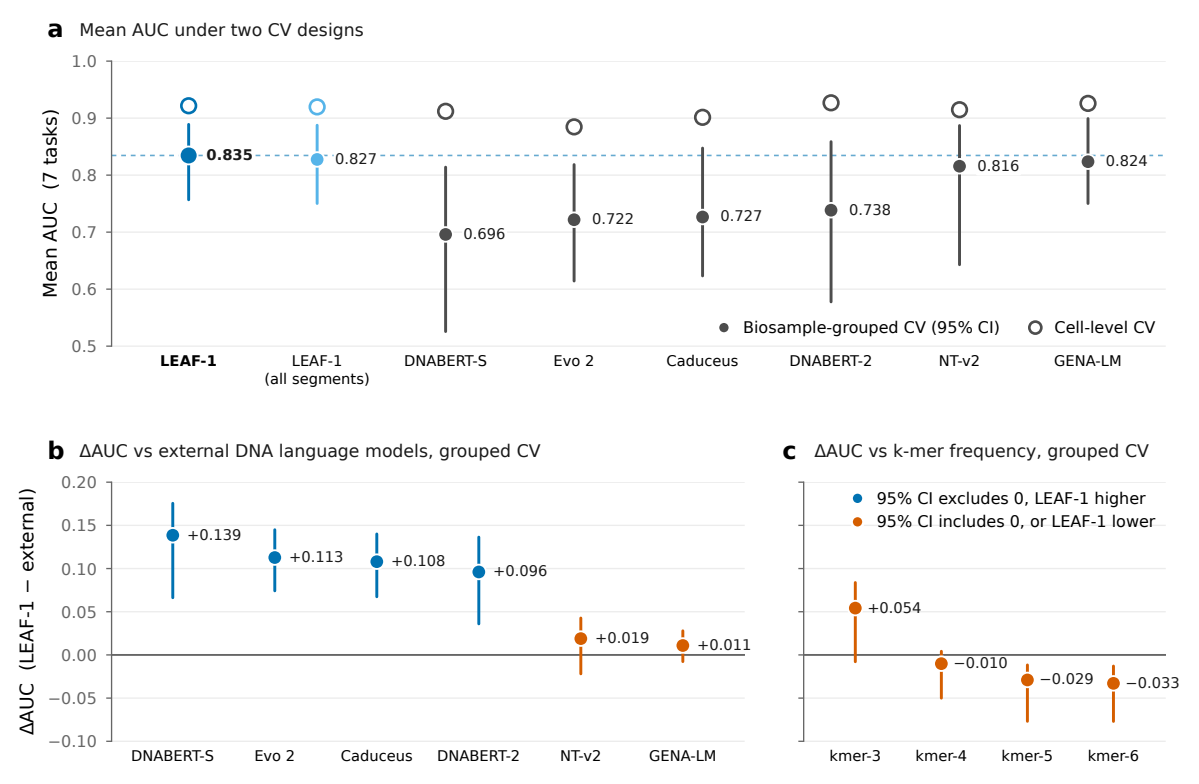
a
Mean AUC under two CV designs
Mean AUC (7 tasks)
1.0
0.9
0.8
0.7
0.6
0.5
0.835
0.827
0.696
0.722
0.727
0.738
0.816
0.824
Biosample-grouped CV (95% CI)
Cell-level CV
LEAF-1
LEAF-1 (all segments)
DNABERT-S
Evo 2
Caduceus
DNABERT-2
NT-v2
GENA-LM
b
ΔAUC vs external DNA language models, grouped CV
ΔAUC (LEAF-1 − external)
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
+0.139
+0.113
+0.108
+0.096
+0.019
+0.011
DNABERT-S
Evo 2
Caduceus
DNABERT-2
NT-v2
GENA-LM
c
ΔAUC vs k-mer frequency, grouped CV
95% CI excludes 0, LEAF-1 higher
95% CI includes 0, or LEAF-1 lower
+0.054
−0.010
−0.029
−0.033
kmer-3
kmer-4
kmer-5
kmer-6